Emerging Role of Cryoneurolysis in Spasticity Management: A Narrative Review

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KEYWORDS

• Cryoneurolysis • Spasticity • Nerve block • Diagnostic motor block • Ultrasound-guided intervention • Neuroplasticity

KEY POINTS

- Cryoneurolysis is an emerging technique with many advantages for managing spasticity in adult and pediatric patients.
- It produces Wallerian degeneration while preserving connective tissue and vessels, minimizing neuroma risk and enabling targeted axonotmesis.
- Clinical evidence, although limited to observational studies and case series, shows long-lasting effects of spasticity reduction across diverse neurologic conditions.
- Adverse effects are mostly mild and temporary, although postprocedural neuropathic pain remains a concern, especially when targeting mixed nerves.
- Cryoneurolysis is potentially cost-effective in select populations and may reduce the frequency of interventions, offering long-term economic benefits.

Video content accompanies this article at http://www.pmr.theclinics.com.

INTRODUCTION AND BACKGROUND

The therapeutic application of cold has a long and well-documented history in medicine. The Edwin Smith Papyrus, dating to around 3500 BC, includes references

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Abbreviations	
BoNT	botulinum toxin
CP	cerebral palsy
MS	multiple sclerosis
SCI	spinal cord injury
TBI	traumatic brain injury

to the use of cold in the treatment of various ailments.[1] Hippocrates, in 400 BC, advocated the use of cold water for its analgesic and anti-inflammatory properties.[2] Dr James Arnott, a UK physician working in the mid-1800s, used a mixture of ice and salt to lower the freezing temperature of water, which he then applied to cancerous tissue to arrest inflammation and relieve pain.[3] In the 1870s, French physicist Louis-Paul Cailletet developed a technique for liquefying gases using the Joule Thompson effect, which refers to the temperature change of a gas or liquid (cryogenic agents or "cryogen") when forced through a narrow orifice.[4] The rapid expansion of the compressed cryogen will then extract heat from the surrounding environment, resulting in a final cooling temperature that will depend on the cryogen's intrinsic caracteristics.[5] At the beginning of the twentieth century, this breakthrough in physics paved the way for the medical use of new cryogens. William Pusey, a Chicago-based physician, helped popularize the use of carbon dioxide snow, publishing an article in *JAMA* in 1907 that demonstrated its effects on a black hairy nevus in a child.[6] Hall-Edwards, a radiologist from the United Kingdom, described a carbon dioxide snow collector in an article in *The Lancet* in 1911.[7] The modern era of cryotherapy really began in 1961, when Irving Cooper, an American surgeon, published results from his use of a liquid nitrogen probe that could produce temperatures of −176°C at the tip to cryoablate the basal ganglia of patients with Parkinson disease.[8] John Lloyd, an Oxford-based pain specialist, pioneered a bedside cryo device for the treatment of peripheral pain syndromes and published a seminal article in *The Lancet* in 1976.[9] Cold has also been shown to temporarily reduce muscle hypertonia and the ratio between maximum H-wave and maximum M-wave in subjects living with spasticity.[10,11] The first case of percutaneous nerve targeting cryotherapy for spasticity reduction was published in 1998.[12] It took more than 20 years for the next article on the topic to appear in 2019.[13]

In this review, the term cryoneurolysis is used to describe this procedure, which the authors define as an ultrasound- and electrical stimulation–guided intervention that results in a cold-induced reversible nerve lesion, aiming to achieve axonotmesis for spasticity management. The authors review the mechanisms of action of cryoneurolysis and current evidence for spasticity management in adult and pediatric populations and discuss future directions.

MECHANISM OF ACTION

Although the precise mechanism of action in human subjects remains incompletely understood, extensive basic science and animal research dating back to the 1980s has explored the microscopic and macroscopic effects of cryoneurolysis. Animal studies have suggested that when a segment of peripheral nerve is frozen at temperatures between −20°C and −100°C, Wallerian degeneration subsequently occurs distally, preserving the integrity of the surrounding endoneurial, perineurial, and epineural structures.[14] This equates to a second-degree nerve injury on the Sunderland classification.[15] This degree of nerve injury avoids risk of neuroma formation and aberrant axon regeneration, which, it has been hypothesized, is a significant advantage compared with phenol or alcohol neurolysis.[14] Interestingly, the investigators describe

that temperatures ranging from +10°C to −20°C tend to induce neuropraxia, whereas temperatures less than −100°C are associated with neurotmesis.[14] For cryoneurolysis to be effective, and to avoid hypothesised risks of neuroma formation or destruction of local structures, a target temperature range of -60 to -100, maintained for a defined duration is recommended[16] (Video 1).

Microscopically, a study by Myers and colleagues[17] in rats showed some margination of white blood cells to the vessel wall, and extravasation of erythrocytes into the endoneurial space 90 minutes after a 1-minute cryolesion at −60°C to sciatic nerves. Between 90 minutes and 6 hours after injury, swollen and hydropic Schwann cells with electron-lucent cytoplasm and reduced numbers of organelles were observed; axonal microtubules and neurofilaments were degenerating, and abnormal vesicular structures appeared within axons, with intact basal membranes. Light microscopy confirmed Wallerian degeneration, and endoneurial fluid pressure associated with it was maximal 6 days after the injury.[17] There was evidence of numerous regenerating sprouts at 11 days.[17] In addition to regeneration of microscopic and macroscopic nerve structures, Shah and colleagues[18] demonstrated recovery of motor function—assessed by dorsiflexion torque—1 month after cryoneurolysis of the common peroneal nerve in rats, accompanied by restoration of axon count.

Several factors have been identified in the literature as critical to the effectiveness and outcomes of cryoneurolysis. One such factor is the temperature of the ice ball, which not only determines the degree of nerve injury but also influences the rate of nerve regeneration. Zhou and colleagues[19] demonstrated this in an experiment involving 18 rabbits, in which lower lesion temperatures were associated with delayed sciatic nerve regeneration.

A laboratory investigation by Said and colleagues[20] explored the various factors influencing the size of the effective cryoneurolysis zone, identifying probe size as the most influential variable, followed by temperature, type of cryogen, and probe shape. The investigators highlight that translating these laboratory findings into clinical practice is limited owing to anatomic and physiologic differences of live human tissue.[20] Importantly, various factors could also influence the temperature of the ice ball.[20] There may well be temperatures of a visualised ice ball that are as high as 0 to -20°C.[20] At these temperatures, the result is neuropraxia rahter than axonotmesis.[14] As such, the result can be partial or incomplete and potentially very short lived. Furthermore, an incomplete cryolesion, insufficient to induce full axonotmesis, may induce hyperalgesia lasting several weeks,[21] not to be confused with neuropathic pain lasting multiple months observed in laboratory animals with complete lesions and concurrent nerve manipulation.[22,23]

The exact percentage of axonal loss to consider a lesion incomplete remains unclear. Although most animal studies, including an immunofluorescence study following up rat sciatic nerves by Hsu and Stevensen,[24] show a significant reduction in axons 2 weeks after cryoneurolysis, which normalizes along with nerve and surrounding soft tissue regeneration over 12 to 16 weeks, they fail to demonstrate axonal degeneration of all axons.[18,24–26]

Despite increasing understanding of the cryoneurolysis mechanism of action, a knowledge gap remains to be addressed to improve cryoneurolysis outcomes in the management of adult and pediatric spasticity.

CURRENT EVIDENCE IN HUMAN STUDIES

Randomized controlled trials are currently underway (ClinicalTrials.gov: NCT06782464[27], NCT06726434[28] and NCT07303582;[29]; both in the recruitment phase);

however, to date, only repeated-measures observational pilot studies and case series have been published reporting the effects of cryoneurolysis on human spasticity management. Fourteen full articles have been published reporting cases of cryoneurolysis for spasticity management. Supplemental Table 1 summarizes the published articles. This review synthesizes findings from 2 studies[30,31] and 12 case series[12,13,32–41] published between 1998 and 2025, encompassing a total of 142 subjects after excluding duplicate patient data reported in 2 publications.[30,31] It highlights the clinical applications, efficacy, and safety profile of cryoneurolysis for the treatment of spasticity.

Indications and Contraindications

Subjects with various pathologic conditions, including but not limited to stroke, multiple sclerosis (MS), traumatic brain injury (TBI), spinal cord injury (SCI), and cerebral palsy (CP) have benefited from cryoneurolysis for spasticity management.

Stroke was the most frequently reported cause of spasticity, affecting 82 patients out of 142 (57.75%). MS was the second most common pathologic condition, with 26 subjects (18.31%), followed by 17 subjects (11.97%) with CP, 7 (4.93%) with SCI, 2 (1.41%) with TBI, 2 (1.41%) with genetic spastic paraparesis, 1 (0.70%) with myelopathy owing to HIV, 1 (0.70%) with anoxic brain injury, and 1 (0.70%) with right-sided spastic hemiparesis from unspecified cause.

All articles reported spasticity refractory to standard management, including botulinum toxin (BoNT) injections, physiotherapy, and splinting as part of the indication for cryoneurolysis. In only 5 cases, BoNT had not been previously administered: one owing to contraindication during pregnancy[13]; one owing to aspiration risk[32]; one patient sought an alternative to BoNT[39]; one required an immediate intervention for a marked increase in baseline spasticity 1 month after total knee replacement, which severely impaired ambulation[39]; and one had contracture-related spasticity and expressed concerns about BoNT use.[39] Finally, a prior positive diagnostic nerve block with lidocaine or bupivacaine was reported in all cases.

Intervention

All articles mentioned using a double guidance technique to confirm precise localization of the targeted nerve. Fluoroscopic guidance paired with electrical nerve stimulation was used in the first case report study[12] only, and all other articles used ultrasound guidance paired with electrical nerve stimulation. This trend is representative of the widespread use of ultrasound in clinical practice. Video 2 illustrates a cryoneurolysis procedure performed in a clinical setting on a patient with spasticity, using combined ultrasound and nerve stimulation guidance.

All but one article[12] specified the type of probe used. The Lloyd Neurostat SL 2000 (Spembly Medical, San Diego, CA, USA), using carbon dioxide to reach a temperature of −60°C, was used in 4 articles,[13,30–32] and the hand-held Iovera (Pacira Bioscience, NJ, USA), designed to achieve a target temperature of −88°C with nitrous oxide, was used in 11 articles.[30,31,33–41]

All articles but two[32,39] reported a subcutaneous anesthetic injection (1% lidocaine) before introducing the cryoprobe through the skin. Only one case reported sedation with nitrous oxide gas (Pronox) and fentanyl for a 14-year-old patient with CP.[38] Twelve articles mentioned the utilization of a 16-gauge angiocatheter to guide the cryoprobe tip, increase ultrasound echogenicity, and protect the skin from frostbite.[13,30–38,40,41]

Targeted nerves were initially limited to specific motor branches, but a wider variety of intramuscular motor branches have been cryoneurolysed. Examples include lateral and medial pectoral nerves to pectoralis major and minor; thoracodorsal nerve to latissimus dorsi; branches of the subscapular nerve to teres major and subscapularis;

branches of the femoral nerve to rectus femoris, vastus intermedius, iliacus, and sartorius; branches of the sciatic nerve to semimembranosus and semitendinosus; branches of the median nerve to flexor digitorum superficialis, flexor carpi radialis, and palmaris longus; branches of the ulnar nerve flexor carpi ulnaris.[36–38,41] Targeted nerve trunks included the musculocutaneous, median, ulnar, obturator, and tibial nerves.[12,31,36,39] The procedure was repeated in 5 case series, 8 to 15 months after the initial cryoneurolysis treatment.[13,35,37,40,41]

Advantages

Cryoneurolysis fills some gaps left by other modalities used for spasticity management. Depending on the country of practice, many muscles treated with BoNT are off-label. Phenol and alcohol do not have an on-label indication for spasticity. They are also typically avoided for larger sensorimotor trunks due to the risk of dysaesthesia, as well as potential tissue destruction and vessel thrombosis.[42] Cryoneurolysis, on the contrary, at temperatures used for spasticity management spares the surrounding tissue and only destroys the axons. It can also be performed with the probe adjacent to blood vessels, as heat from blood flow helps protect their integrity.[43,44] Another advantage of cryoneurolysis is the absence of a maximal dose, which allows more muscles to be treated, including all muscles supplied by a nerve trunk. Targeting the ubiquitous spastic claw-hand deformity is now possible via the sensorimotor ulnar and median nerve trunks to address contracture and pain.[42] In addition, cryoneurolysis is a biophysical phenomenon, meaning it does not trigger allergic reactions or antibody formation. Another advantage of cryoneurolysis is its immediate effect after the intervention, with no waiting period between sessions. This allows for a stepwise therapeutic approach—beginning with the most relevant neuromuscular targets in a single session. The patient, caregivers, and rehabilitation team can then assess the initial results and, if necessary, escalate the treatment strategy with additional sessions.

Adverse Events and Side Effects

Periprocedural discomfort is not uncommon with cryoneurolysis. Although frequently observed in clinical practice, it has been reported in only 4 cases in the authors' literature review. One case with poststroke spasticity experienced light-headedness and nausea that recovered quickly after the procedure.[33] One case with MS had tolerable burning pain during the procedure and mild residual pain and bruising around the injection site immediately after.[34] One subject with CP reported cramping and burning pain during the procedure, managed with fentanyl administration, a rash associated with procedures, and agitation following procedural sedation.[38] One case diagnosed with MS experienced severe cramping during the brachialis treatment, which resolved with heat and massage.[40] Regarding postprocedural pain, a recent case series reported that 6 of 8 patients experienced discomfort following the intervention; in all but one case, symptoms resolved within 1 month.[41] For the sixth patient, pain had fully dissipated by the 3-month follow-up.[41] Analysis of more than 100 children who have undergone cryoneurolysis for spasticity is ongoing. For most pediatric patients, the procedure is done under full general sedation, although some children have tolerated the procedure awake. Other analgesic treatments, including nitrous oxide gas, methoxyflurane (Penthrox), oral medications, sedatives, and topical preparations, can be offered to patients.[42]

Prolonged postprocedural reported adverse events include the following:

- One case of skin infection treated with antibiotics[30];
- Five cases of bruising or swelling[30,34,41];

- Fifteen cases of persistent nerve pain or dysesthesia treated with topical lidocaine, nonsteroidal anti-inflammatory drugs, lidocaine and cortisone injection, gabapentinoids, or BoNT injection[30,41];
- One case of cramping in the antagonist muscle treated with BoNT.[30]

An article analyzing the adverse effects of cryoneurolysis for spasticity treatment found that treating nerve trunks can lead to temporary neuropathic pain and/or dysesthesia, despite the theoretic prevention of neuromas through preservation of neuronal structural connective tissue.[30] Based on animal models, neurogenic pain could be due to an incomplete cryolesion[21] or neuropathic pain with complete lesions and concurrent nerve manipulation.[22,23] Still, no studies have yet explored the pathophysiology of postcryoneurolysis pain in humans. Postcryoneurolysis nerve pain has also been observed after targeting intramuscular motor branches, but is much more common in mixed nerves, with 12 of 15 cases of nerve pain or dysesthesia occurring in sensorimotor nerve trunks.[30,41]

In clinical practice, postprocedural muscle pain also occurs, hypothetically explained by muscle stretching and soreness owing to increased passive and sometimes active range of motion, as well as occasional functional movements inaccessible before cryoneurolysis. Other potential adverse events include muscle weakness and functional loss from treating too many muscles. In addition, although conditions such as adhesive glenohumeral capsulitis, joint instability, or antagonist muscle contractures may already be present, they are often masked by spasticity-related joint stiffness. When spasticity is successfully treated, these underlying musculoskeletal issues can become clinically apparent. This outlines the importance of a good postintervention follow-up and pain generator identification.

Patient Satisfaction

Despite the risk of adverse events and side effects, reported qualitative patients' experiences in 4 articles[33–35,41] show that patients' satisfaction is generally high, as follows:

- "It is now easier to get dressed and easier to wash under my arm. [Since the procedure] I have no pain whereas I used to have pain especially when I got tired."[33]
- "Now I can go all over the playgrounds and the parks with my grandkids. I can walk several kilometers, I do get tired, but I'm not in pain anymore."[33]
- "I am delighted. It's an improvement in quality of life which is critical for people like me. Do you know how liberating it is to be able to shave your underarm? It's a big step for me."[33]
- "Any pain I experienced was completely overshadowed by how amazing I felt right after."[34]
- "Almost immediately, when I got off the bed, it was clear that the pain had improved right away, and I had increased mobility. We all had a hug and a cry. It's just remarkable this treatment really. I lifted my toes up off the floor for the first time in 20 years."[34]
- "As the weeks passed, I had even more increased mobility and decreased pain. At Christmas, I got to show my friends and family my new party trick where I get up off the floor by extending my toes and pushing into the ground. I hadn't been able to do this for 20 years"[34]

A case series of 8 adults by Pick and colleagues[41] asked all subjects to complete a patient satisfaction questionnaire, including a visual analogue scale to rate their overall satisfaction with the treatment and a question about whether they would recommend

the treatment to others.[41] At 6 months, 7 participants rated their satisfaction as 10 out of 10; 1 rated it as 9 out of 10, and 7 stated they would recommend the treatment to others. At 12 months, 4 out of 5 participants who completed the questionnaire rated their satisfaction as 10 out of 10, one rated it as 8 out of 10, and they all stated they would recommend the treatment to others.[41]

Cost Considerations

A 2025 case report compared the cost of BoNT injections to cryoneurolysis in a patient on a high dose of 600 units of onabotulinumtoxinA 4 times a year, representing a value of at least 10,798.44 USD.[40] The patient received 2 treatments of cryoneurolysis in 2 years at a cost of 460 USD each cycle, plus the initial cost of reusable equipment, and did not require further treatments.[40] A review of 77 patients treated in a Canadian province found that over 18 months, the cost of spasticity management health care for the province was reduced by 123 CAD per patient per month on average (95% CI: $109–$137).[45] However, these cost-savings estimates highlight the importance of remaining cautious until a more definitive health economic investigation is carried out.[45]

DISCUSSION AND FUTURE CONSIDERATIONS

Cryoneurolysis is a promising tool for managing spasticity. The current literature review indicates a clear trend toward its significant potential in both adult and pediatric spasticity management. However, some questions regarding its mechanism of action in adults and children living with spasticity remain. Animal models help enhance the understanding, but they are not entirely translatable to humans. Moreover, to determine the role of cryoneurolysis in the multimodal spasticity treatment approach, future research should answer some questions.

What Are the Ideal Cryoneurolysis Parameters, and How Should We Control Them in Adults and Children?

Parameters, such as the volume of biological tissue exposed to cryogenic temperatures, the spatial extent of exposure, and the duration of the freezing and thawing, are critical parameters influencing treatment outcomes.[16,19,20] The temperature of the targeted tissues appears to differ from the values predicted based on the properties of the cryogen employed,[20] which could significantly impact the duration of the effect[19] and elevate the risk of side effects like neuropathic pain associated with partial neurolysis.[21] The ice ball itself is a visible but indirect indicator of thermal spread.[16] Therefore, attention should be given to the temperature gradient established within the targeted tissue over time, ranging from body temperature to the lowest temperatures achieved.[16] Within this gradient, portions of tissue may experience neuropraxic effects (approximately $+10^{\circ}$C to -20°C), whereas others may reach thresholds necessary for axonotmesis or neurotmesis.[16] This complex spatiotemporal interaction demands further investigation to optimize safety and efficacy, as highlighted in recent findings.[16]

How to Explain the Long-Term Response to Cryoneurolysis in Spastic Patients?

Beyond the Wallerian degeneration and the full recovery of neural and muscular tissues reported to occur 12 to 16 weeks after the insult in rodens,[17,18,24] the long-term effects of cryoneurolysis are poorly understood. Clinical benefits have been noted for a much longer period in many publications.[12,13,31,32,34,35,37,38,40,41] Another surprising finding in the outcomes of many reported cases is the increment in V1 of the Tardieu assessment following cryoneurolysis, indicative of increased passive range of motion.[31,35–38,40] The authors hypothesize that the intense spasticity reduction following cryoneurolysis

provides a unique therapeutic window, allowing intense stretching that was previously impossible. The loss of sensory afferences[46] and cerebrospinal neuroplasticity might also contribute to those results. Those hypotheses still need to be proven. Whether the duration from animal studies can be translated to humans also remains unknown.

What Nervous Structure Should Be Preferably Targeted by Cryoneurolysis?

Theoretically, aiming for a sensorimotor nerve trunk rather than a motor branch does not yield the same result. However, current literature fails to explore the difference in outcome, including the degree and duration of the improvement and the development of adverse effects, with respect to this key variable. It is suggested that targeting motor branches causes less dysesthesia and neuropathic pain, but it can still occur.[30] This might be explained by the presence of sensory afferent fibers within "pure motor" nerves.[46] Following the same logic, motor endplates could also be a target. Although it might be tempting to think that motor branches or endplates should be preferred, it is unclear if the effects would be as significant or long lasting. It must also be remembered that more entry points and individual lesions are required when targeting motor branches or endplates. This can considerably increase the total procedure duration and challenge patients' tolerability.

How Should the Periprocedural Pain Be Managed?

Pericryoneurolysis pain was the most reported adverse event in this review. In one case series, patients received a diagnostic nerve block with local anesthetic on the same day as the cryoneurolysis procedure, limiting pain during the procedure, but removing the potential to stimulate the target with the cryoprobe to confirm optimal positioning.[41] Sedation and analgesia with nitrous oxide gas and fentanyl have been used in one case report.[38] General anesthetic is offered only in some settings, particularly in children or adults with difficult to control movements. Some clinicians also use other pain management strategies like vibration therapy and virtual reality. The benefits and efficacy of these options should be studied in future work.

How Frequent and Debilitating Is the Postprocedure Pain, How Long Does It Last, and How Should It be Addressed?

Although neuropathic pain has been the most commonly reported persistent adverse effect,[30,41] soft tissue–related pain, or an underlying musculoskeletal condition, could also contribute to postprocedure pain. The pain generator must be determined before considering appropriate management. To prevent neurogenic pain, some clinicians suggest increasing the number and/or duration of cycles, especially for larger nerves, to avoid incomplete axonotmesis. As discussed previously, target selection may also impact the risk of postprocedure pain. The reviewed articles describe various options to manage postprocedural pain: topical lidocaine, nonsteroidal anti-inflammatory drugs, lidocaine, cortisone, or BoNT injections, and gabapentinoids, which were by far the most frequently used strategy.[30,41] However, no study has focused on the cause and best management of postcryoneulorysis pain.

Who Is a Suitable Patient for Cryoneurolysis?

Patient selection remains a crucial aspect that needs clarification. Because of the potentially long-lasting effects, it seems reasonable to reserve cryoneurolysis for a population that has plateaued with usual management. However, the question of cryoneurolysis as an early intervention has been raised by many experts and warrants further study. Positive outcomes have been reported with patients suffering from various pathologic conditions causing spasticity, including stroke, MS, CP, SCI, TBI,

and anoxic brain injury. Questions regarding which populations would benefit the most from cryoneurolysis, and variations that should be made to the protocol according to the underlying causal pathologic condition, should be elucidated.

Is Cryoneurolysis a Financially Suitable Treatment for Spasticity Management?

Based on currently available data,[40,45] it is reasonable to think that cryoneurolysis provides monetary benefits for the management of spasticity in some patients. As cryoneurolysis becomes more integrated into routine care and the number of patients managed with this approach grows—benefiting from longer-lasting effects and fewer interventions—the economic impact is expected to increase further. Larger comprehensive cost-analysis studies that account for local resource availability across diverse health care settings are warranted, particularly given the current context of widespread financial constraints in many health systems.

What Is the Role of Cryoneurolysis in the Multimodal Management Approach?

It is widely accepted that a multimodal approach should be privileged to address the different components of spastic movement disorders.[47] Current evidence suggests that cryoneurolysis can have a key role in the future of spasticity management, but its exact place in the multimodal treatment algorithm remains to be further examined.

CLINICS CARE POINTS

- Cryoneurolysis is a minimally invasive technique showing growing promise in the management of focal spasticity.
- This review synthesizes evidence on the mechanism of action, procedural considerations, and clinical outcomes across diverse neurologic conditions.
- Preliminary data demonstrate efficacy in reducing tone and improving function, but high-quality randomized studies are still awaited.
- Nerve and nerve branch target selection, timing relative to other interventions, and standardized outcome measures are critical to optimizing efficacy.
- Further research is needed to empirically demonstrate efficacy, refine protocols, and clarify the role of cryoneurolysis within comprehensive spasticity management paradigms.

DISCLOSURE

Dr È. Boissonnault has received honoraria for consultant services from AbbVie, Ipsen, Merz, and Pacira, an educational grant from AbbVie, and served on advisory boards for Ipsen and Merz. Dr A. Pick has received teaching fees and honoraria from Merz and Ipsen. Dr J. Pereira has received travel support for conference participation from Metrum CryoFlex, AbbVie, Ipsen, Medtronic, and Merz and serves as a Merz Benelux Advisory Board member. Dr S. Li has received consultant honoraria and research grants from Pacira, a research grant from Saol Therapeutics, and consultant honoraria from Abbvie and Merz. Dr P. Winston has received honoraria and educational and research grants and served on advisory boards for AbbVie, Ipsen, Merz, and Pacira.

SUPPLEMENTARY DATA

Supplementary data to this article can be found online at https://doi.org/10.1016/j.pmr.2026.01.009.

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